

Chapter 2 - Sheaves

Flandre

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1 Motivating example: The sheaf of smooth functions

Exercise 1.A

Show that this is the only maximal ideal of O_p . (Hint: show that every element of $O_p \setminus \mathfrak{m}_p$ is invertible.)

Proof. For $\forall (f, U) \in \mathcal{O}_p \setminus \mathfrak{m}_p$, we have

$$\left(\frac{1}{f}, U'\right) \in \mathcal{O}_p \setminus \mathfrak{m}_p$$

in a neighbourhood U' of p , hence invertible. □

Exercise 1.B [Skipped. I haven't learnt Differential Geometry yet]

EXERCISE FOR THOSE WITH DIFFERENTIAL GEOMETRIC BACKGROUND.

Prove this. (Rhetorical question for experts: what goes wrong if the sheaf of continuous functions is substituted for the sheaf of smooth functions? What goes wrong if you use the sheaf of C^1 functions?)

2 Definition of sheaf and presheaf

Exercise 2.A [Trivial]

EXERCISE FOR CATEGORY-LOVERS: "A PRESHEAF IS THE SAME AS A CONTRAVARIANT FUNCTOR".

Given any topological space X , we have a "category of open sets" (Example 1.1.9), where the objects are the open sets and the morphisms are inclusions. Verify that the data of a presheaf is precisely the data of a contravariant functor from the category of open sets of X to the category of sets. (This interpretation is surprisingly useful.)

Exercise 2.B

UNIMPORTANT EXERCISE: PRESHEAVES THAT ARE NOT SHEAVES.

Show that the following are presheaves on $\mathbb{C}$ (with the classical topology), but not sheaves: (a) bounded functions, (b) holomorphic functions admitting a holomorphic square root.

Proof. (a) Define

$$\mathcal{F}(U) = \{f : U \longrightarrow \mathbb{C} \mid f \text{ is bounded in } U\}$$

Obviously $\mathcal{F}$ forms a presheaf.

Now consider

$$U_n = \{z \in \mathbb{C} \mid |z| \leq n\}, n = 1, 2, 3, \dots$$

and define $f_n \in \mathcal{F}(U_n)$ by

$$f_n(z) = z$$

obviously the corresponding glued f cannot be bounded, hence not in $\mathcal{F}$.

(b) (Skipped. I've forgott the theories in complex analysis.) □

Exercise 2.C

The identity and gluing axioms may be interpreted as saying that $\mathcal{F}(\cup_{i \in I} U_i)$ is a certain limit. What is that limit?

Solution. It's $\mathcal{F}(U)$.

Exercise 2.D [Trivial]

- (a) Verify that the examples of §2.1 are indeed sheaves (of smooth functions, or continuous functions, or real-analytic functions, or plain real-valued functions, on a manifold or $\mathbb{R}^n$).
- (b) Show that real-valued continuous functions on (open sets of) a topological space X form a sheaf.

Exercise 2.E

(CONSTANT SHEAVES).

Now let $\mathcal{F}(U)$ be the set of maps $U \rightarrow S$ that are locally constant, i.e., for any point p in U , there is an open neighborhood of p where the function is constant. Show that this is a sheaf. (A better description is this: endow S with the discrete topology, and let $\mathcal{F}(U)$ be the continuous maps $U \rightarrow S$.) This is called the constant sheaf (with values in S); do not confuse it with the constant presheaf. (I would prefer the name "locally constant sheaf", but it is too late in history for this change.) We denote this sheaf $\underline{S}$.

Proof. (I suppose it's trivial, because the gluing axiom is satisfied in this case.) □

Exercise 2.F [Trivial]

("MORPHISMS GLUE").

Suppose Y is a topological space. Show that "continuous maps to Y " form a sheaf of sets on X . More precisely, to each open set U of X , we associate the set of continuous maps of U to Y . Show that this forms a sheaf. (Exercise 2.2.D(b), with $Y = \mathbb{R}$, and Exercise 2.2.E, with $Y = S$ with the discrete topology, are both special cases.)

Exercise 2.G

This is a fancier version of the previous exercise.

(a) (sheaf of sections of a map) Suppose we are given a continuous map $\mu : Y \rightarrow X$. Show that "sections of μ " form a sheaf. More precisely, to each open set U of X , associate the set of continuous maps $s : U \rightarrow Y$ such that $\mu \circ s = \text{id}_U$. Show that this forms a sheaf. (For those who have heard of vector bundles, these are a good example.) This is motivation for the phrase "section of a sheaf".

(b) (This exercise is for those who know what a topological group is. If you don't know what a topological group is, you might be able to guess.) Suppose that Y is a topological group. Show that continuous maps to Y form a sheaf of groups.

Proof. (a) (Trivial).

(b) (The same as **Exercise 2.2.F**.) □

Exercise 2.H [Trivial]

IMPORTANT EXERCISE/DEFINITION: THE PUSHFORWARD SHEAF OR DIRECT IMAGE SHEAF.

Suppose $\pi : X \rightarrow Y$ is a continuous map, and F is a presheaf on X . Then define a presheaf $\pi_* F$ on Y by $\pi_* F(V) = \mathcal{F}(\pi^{-1}(V))$, where V is an open subset of Y . Show that $\pi_* F$ is a presheaf on Y , and is a sheaf if F is. This is called the pushforward (or direct image) of F . More precisely, $\pi_* F$ is called the pushforward of F by π .

Exercise 2.I

(PUSHFORWARD INDUCES MAPS OF STALKS).

Suppose $\pi : X \rightarrow Y$ is a continuous map, and $\mathcal{F}$ is a sheaf of sets (or rings or λ -modules) on X . If $\pi(p) = q$, describe the natural morphism of stalks $(\pi_* \mathcal{F})_q \rightarrow \mathcal{F}_p$. (You can use the explicit definition of stalk using representatives, §2.2.4, or the universal property, §2.2.5. If you prefer one way, you should try the other.)

Solution. Notice

$$\begin{array}{ccc}
 & \mathcal{F}_p & \\
 & \uparrow \exists! & \\
 & (\pi_* \mathcal{F})_q & \\
 \nearrow & \uparrow & \nearrow \\
 \pi_* \mathcal{F}(\tilde{V}) & \longrightarrow & \pi_* \mathcal{F}(\tilde{U})
 \end{array}
 \qquad
 \begin{array}{ccc}
 & \mathcal{F}_p & \\
 & \uparrow & \\
 \mathcal{F}(V) & \longrightarrow & \mathcal{F}(U)
 \end{array}
 .$$

Done.

Exercise 2.J

If $(X, \mathcal{O}_X)$ is a ringed space, and F is an $\mathcal{O}_X$ -module, describe how for each $p \in X$, F_p is an $\mathcal{O}_{X,p}$ -module.

Solution. (Use 2.2.4. directly).

3 Morphisms of presheaves and sheaves

Exercise 3.A

MORPHISMS OF (PRE)SHEAVES INDUCE MORPHISMS OF STALKS.

If $\phi: \mathcal{F} \rightarrow \mathcal{G}$ is a morphism of presheaves on X , and $p \in X$, describe an induced morphism of stalks $\phi_p: \mathcal{F}_p \rightarrow \mathcal{G}_p$. Translation: taking the stalk at p induces a stalkification functor $\text{Sets}_X \rightarrow \text{Sets}$. (Your proof will extend in obvious ways. For example, if ϕ is a morphism of $\mathcal{O}_X$ -modules, then ϕ_p is a map of $\mathcal{O}_{X,p}$ -modules.)

Solution. We know elements of stalk $\mathcal{F}_p$ consists of (f, U) where $p \in U, f \in \mathcal{F}(U)$ and we have $(f, U) \sim (g, V)$ iff

$$\exists W \subseteq U \cap V, f|_W = g|_W.$$

Now, consider their images under the presheave morphism :

$$\phi: \mathcal{F} \longrightarrow \mathcal{G}$$

$$\phi_U: (f, U) \longmapsto (\phi(f), U)$$

where $(\phi(f), U) \in \mathcal{G}_p$. We need to show the induced map

$$\phi_p: \mathcal{F}_p \longrightarrow \mathcal{G}_p$$

is well-defined.

For $(f, U) \sim (g, V)$ where $f|_W = g|_W$, we have

$$\begin{aligned}
 \phi_U(f)|_W &= \phi_W(f|_W) \\
 &= \phi_W(g|_W) \\
 &= \phi_V(g)|_W
 \end{aligned}$$

Done.

Exercise 3.B [Trivial]

Suppose $\pi: X \rightarrow Y$ is a continuous map of topological spaces (i.e., a morphism in the category of topological spaces). Show that pushforward gives a functor $\pi_*: \text{Sets}_X \rightarrow \text{Sets}_Y$. Here Sets can be replaced by other categories. (Watch out for some possible confusion: a presheaf is a functor, and presheaves form a category. It may be best to forget that presheaves are functors for now.)

Exercise 3.C [Trivial]

IMPORTANT EXERCISE AND DEFINITION: "SHEAF Hom".

Suppose $\mathcal{F}$ and $\mathcal{G}$ are two sheaves of sets on X . (In fact, it will suffice that $\mathcal{F}$ is a presheaf.) Let $\text{Hom}(\mathcal{F}, \mathcal{G})$ be the collection of data

$$\text{Hom}(\mathcal{F}, \mathcal{G})(U) := \text{Mor}(\mathcal{F}|_U, \mathcal{G}|_U).$$

(Recall the notation $\mathcal{F}|_U$, the restriction of the sheaf to the open set U , Example 2.2.8.) Show that this is a sheaf of sets on X . (To avoid a common confusion: the right side does not say $\text{Mor}(\mathcal{F}(U), \mathcal{G}(U))$.) This sheaf is called "sheaf Hom". (Strictly speaking, we should reserve Hom for when we are in an additive category, so this should possibly be called "sheaf Mor". But the terminology "sheaf Hom" is too established to uproot.) It will be clear from your construction that, like Hom , Hom is a contravariant functor in its first argument and a covariant functor in its second argument.

Proof. (Use the property of sheaf $\mathcal{G}$ to establish the sheaf properties in $\text{Hom}(\mathcal{F}, \mathcal{G})$.) $\square$

Exercise 3.D

UNIMPORTANT EXERCISE (REALITY CHECK).

(a) If $\mathcal{F}$ is a sheaf of sets on X , then show that $\text{Hom}(\{p\}, \mathcal{F}) \cong \mathcal{F}$, where $\{p\}$ is the constant sheaf "with values in the one element set $\{p\}$ ".

(b) If $\mathcal{F}$ is a sheaf of abelian groups on X , then show that $\text{Hom}_{\text{Ab}_X}(\mathbb{Z}, \mathcal{F}) \cong \mathcal{F}$ (an isomorphism of sheaves of abelian groups).

(c) If $\mathcal{F}$ is an $\mathcal{O}_X$ -module, then show that $\text{Hom}_{\text{Mod}_{\mathcal{O}_X}}(\mathcal{O}_X, \mathcal{F}) \cong \mathcal{F}$ (an isomorphism of $\mathcal{O}_X$ -modules).

Proof. (Trivial. If not, follow the hint and you'll think it is.) $\square$

Exercise 3.E [Trivial]

Show that $\ker_{\text{pre}} \phi$ is a presheaf. (Hint: if $U \hookrightarrow V$, define the restriction map by chasing the following diagram:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \ker_{\text{pre}} \phi(V) & \longrightarrow & \mathcal{F}(V) & \longrightarrow & \mathcal{G}(V) \\ & & \downarrow \exists! & & \downarrow \text{Res}_{V,U} & & \downarrow \text{Res}_{V,U} \\ 0 & \longrightarrow & \ker_{\text{pre}} \phi(U) & \longrightarrow & \mathcal{F}(U) & \longrightarrow & \mathcal{G}(U) \end{array}$$

You should check that the restriction maps compose as desired.)

Proof. (Just chase the given diagram, trivial.) $\square$

Exercise 3.F

THE COKERNEL DESERVES ITS NAME.

Show that the presheaf cokernel satisfies the universal property of cokernels (Definition 1.5.4) in the category of presheaves.

Proof. By "the universal property of cokernels", we mean there exists unique map

$$\begin{array}{ccccc} & & & \mathcal{H} & \\ & \nearrow 0 & & \uparrow \exists! & \\ \mathcal{F} & \longrightarrow & \mathcal{G} & \longrightarrow & \text{Coker}_{\text{pre}} \phi \end{array}.$$

Specifically, it means that for any $V \subseteq X$ open, there exists unique map

$$\begin{array}{ccccc} & & & \mathcal{H}(V) & \\ & \nearrow 0 & & \uparrow \exists! & \\ \mathcal{F}(V) & \longrightarrow & \mathcal{G}(V) & \longrightarrow & \text{Coker}_{\text{pre}} \phi(V) \end{array},$$

which is already given by the universal property of cokernel $\text{Coker}_{\text{pre}} \phi$ in category of abelian group. $\square$

Exercise 3.G [Trivial]

EASY EXERCISE.

Show (or observe) that for a topological space X with open set U , $\mathcal{F} \mapsto \mathcal{F}(U)$ gives a functor from presheaves of abelian groups on X , Ab_X^{pre} , to abelian groups, Ab . Then show that this functor is exact.

Proof. Chase the following diagram, and the result can be obtained immediately.

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathcal{F} & \longrightarrow & \mathcal{G} & \longrightarrow & \mathcal{H} & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \mathcal{F}(U) & \longrightarrow & \mathcal{G}(U) & \longrightarrow & \mathcal{H}(U) & \longrightarrow & 0 \end{array}$$

□

Exercise 3.H

Show that a sequence of presheaves $0 \rightarrow F_1 \rightarrow F_2 \rightarrow \cdots \rightarrow F_n \rightarrow 0$ is exact if and only if $0 \rightarrow \mathcal{F}_1(U) \rightarrow \mathcal{F}_2(U) \rightarrow \cdots \rightarrow \mathcal{F}_n(U) \rightarrow 0$ is exact for all U .

Proof. (i) ($\Rightarrow$): Given by **Exercise 2.3.G.**, i.e. $(-)(U)$ is an exact functor.

(ii) ($\Leftarrow$): (Use the definition of presheaf kernels and presheaf images and break the sequence of presheaves into sequences of abelian groups.)

□